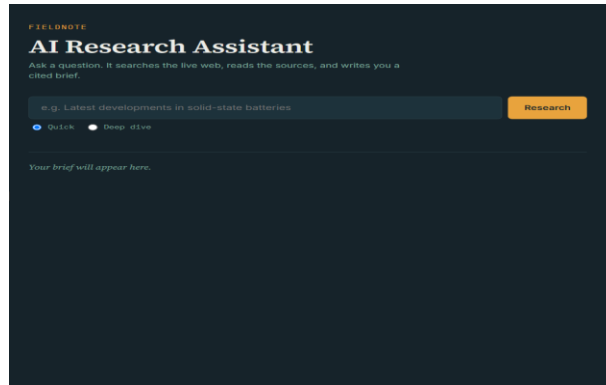


Prompt Ladder Assignment

Version 0 – Baseline

Prompt

Build an AI Research Assistant.



Representative Output

Single HTML research assistant with web search and cited answers.

Reflection

- What changed: Baseline prompt.
- What improved: N/A
- What still failed: Too broad, generic, no audience or constraints.
- What I'll try next: Add a clear goal.

Version 1 – Clear Goal

Prompt

Build an AI Research Assistant that allows users to upload PDFs and ask questions.

Representative Output

PDF-based Q&A assistant grounded in uploaded documents.

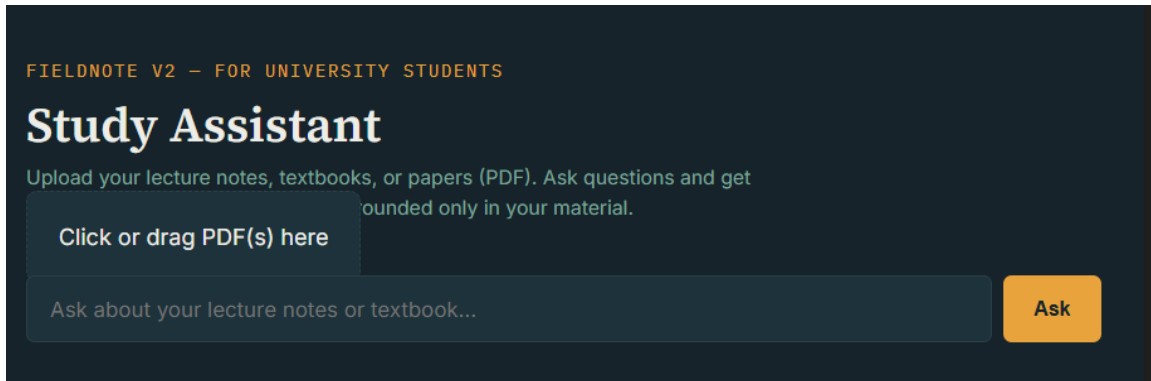
Reflection

- What changed: Added a clear goal.
- What improved: Responses became focused on document Q&A.
- What still failed: No audience or study-focused features.
- What I'll try next: Add target audience.

Version 2 – Audience

Prompt

...for university students...



Representative Output

Student-focused explanations and UI.

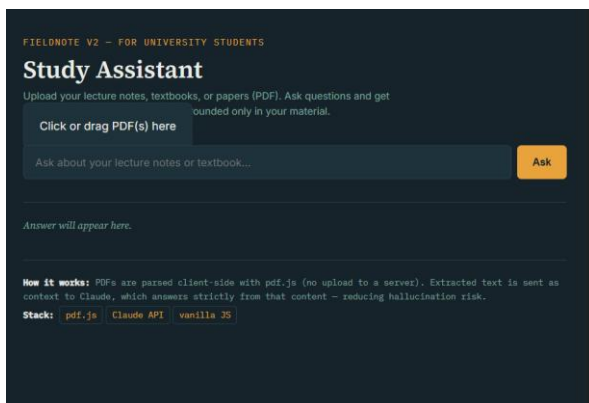
Reflection

- What changed: Added audience.
- What improved: Simpler, exam-friendly responses.
- What still failed: No flashcards or citations.
- What I'll try next: Add context.

Version 3 – Context

Prompt

...for my AI internship portfolio...



Representative Output

Portfolio-oriented documentation and validation.

Reflection

- What changed: Added context.
- What improved: Better reviewer experience and technical reasoning.
- What still failed: Needs README/demo assets.
- What I'll try next: Specify output format.

Version 4 – Output Format

Prompt

Use fixed sections.

Representative Output

Structured seven-section output.

Reflection

- What changed: Added output format.
- What improved: Much easier to read and compare.
- What still failed: Less flexible than free-form explanations.
- What I'll try next: Add quality criteria.

Version 5 – Quality Criteria

Prompt

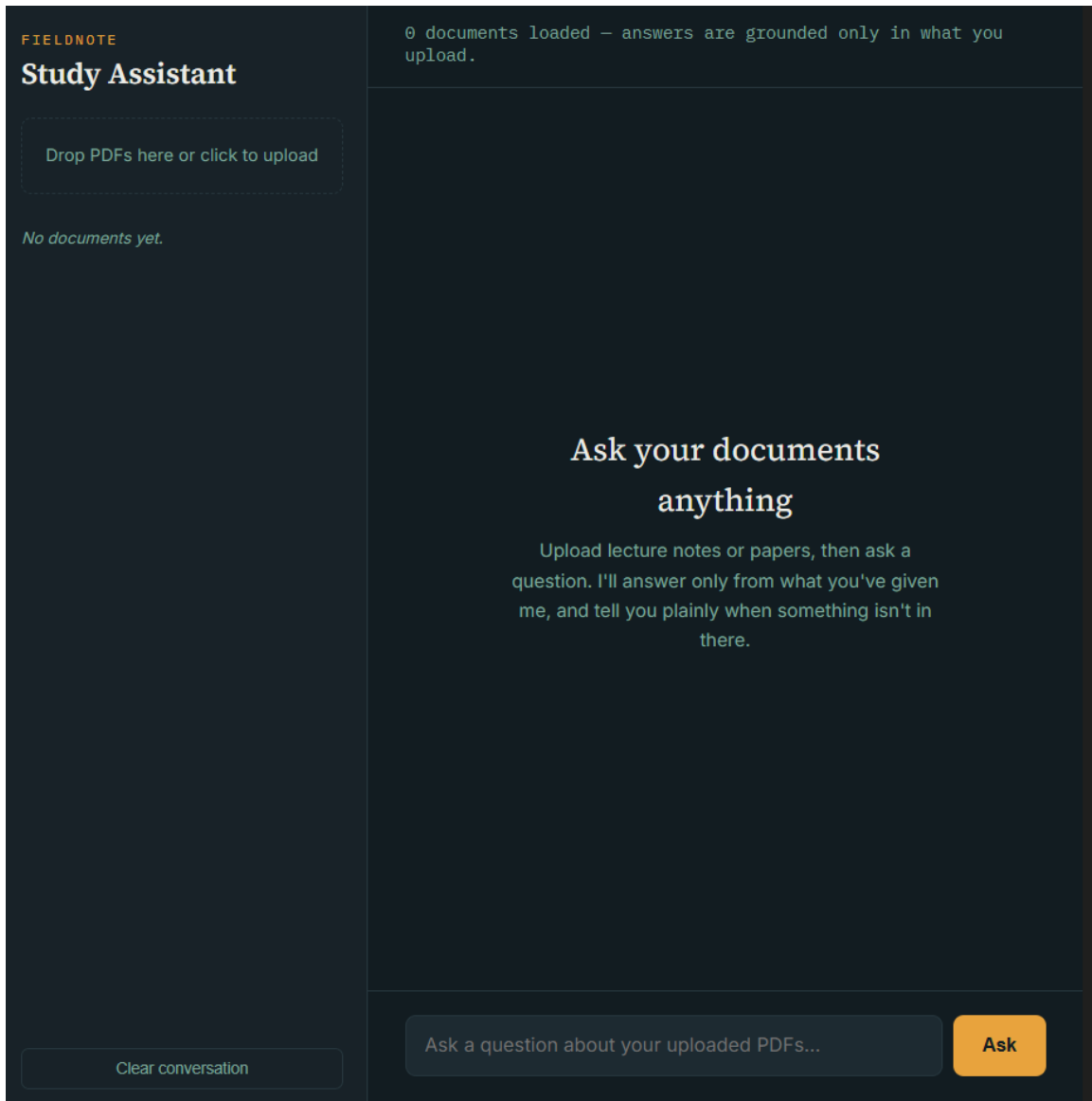
Recommend production-ready architecture.

Representative Output

Production-ready RAG architecture and justified stack.

Reflection

- What changed: Added quality criteria.
- What improved: More practical and implementation-focused.
- What still failed: Still requires coding to implement.
- What I'll try next: Use as final reusable prompt.



Final Reusable Prompt

Build an AI Research Assistant for university students for an AI internship portfolio. Support PDF upload and grounded Q&A using a production-ready RAG architecture. Present the response as: Project Overview, Features, Tech Stack, System Architecture, Folder Structure, Development Steps, Future Improvements. Explain technical decisions, avoid unnecessary complexity, and recommend scalable solutions.